Convergent Event Knowledge for a Common Tactical Picture over Degraded Networks

Brief topic description for Program Officer discussion · FY27 ONR Young Investigator Program (N0001426SF004)

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Problem. Language-model assistants can turn text reports (messages, chat, intelligence summaries) into an event picture: who did what, where, and when. Over degraded links, the same reports arrive late, duplicated, out of order, or not at all, and each node's assistant extracts events from whatever it has seen. Nodes can disagree even after every report has arrived. A destroyer receives "unidentified vessel loitering near the channel entrance as of 0600," then the update "vessel departed the channel area at 0630." A shore node receives the update first and the delayed original second; if the latest arrival wins, its picture shows the vessel still loitering. Event extraction typically assumes one central collection, and data-synchronization methods assume clean symbolic records; neither handles knowledge that uncertain models extract from text.

Central hypothesis. Representing extracted events as small, source-stamped updates with an order-independent merge (commutative, associative, and idempotent) will make nodes that hold the same extracted updates converge to the same picture in any arrival order, and will limit divergence when extraction is uncertain. *Question-directed sharing*, which sends first the updates that change answers to the commander's standing information requirements, will deliver correct answers sooner under bandwidth limits than periodic synchronization.

Research thrusts.

- **Mergeable event updates from text.** Extract each report into updates that record the actor, action, time, place, source, and claim status. Represent corrections and retractions as explicit relations rather than overwrites, and represent uncertain coreference and time normalization as revisable links that keep alternatives until evidence resolves them.
- **Order-independent fusion under extraction uncertainty.** Design merge rules with these three properties for the symbolic event layer, then measure where model ambiguity (coreference, time expressions, paraphrase) breaks convergence. Because a model's reading of a report depends on what it has already read, two nodes can extract the same report differently; we will compare context-independent extraction with reconciliation at merge time and bound the remaining divergence.
- **Question-directed sharing under limited bandwidth.** Express commander's critical information requirements (CCIRs) as standing questions over the event graph. Send pending updates in order of how much they change an answer or its uncertainty. Each answer cites its supporting reports and what may still be missing. Our question-answering formulation of event extraction suggests that one model could both read reports and answer these questions.

What is new. We make language-derived event knowledge mergeable, measure how far extraction uncertainty keeps nodes apart, and decide what to send first.

Evaluation. Event-annotated documents (MAVEN-ERE) and our MEQA questions, replayed across simulated nodes with message delay, loss, duplication, and reordering. Measures: cross-node agreement of event graphs, answer accuracy over time, bytes transmitted, and time to convergence after a link is restored. Baselines: centralized extraction (an upper bound), last-writer-wins updates, periodic full synchronization, and retrieval-augmented generation over each node's local messages.

Naval relevance. Distributed maritime operations must maintain a common tactical picture across ships, aircraft, and shore commands whose links are often disconnected, intermittent, or bandwidth-limited. When links return, nodes should agree without re-sending everything; over a narrow link, the report that changes a CCIR answer should go first. The topic addresses the program's interests in developing a common tactical picture, automated information integration across sensors and information stores, and decision making with uncertain and out-of-sequence information.

PI foundation. Event extraction by answering questions (EMNLP 2020); template filling with generative transformers (NAACL 2021); consistent event–event temporal relation extraction (MATCHING 2023); MEQA, multi-hop event-centric question answering (NeurIPS 2024). The PI's NSF CAREER award (2024) studies extracting consistent event graphs from long, complex documents; the distributed, order-independent setting here is new. As a postdoc, the PI led system integration for the RESIN team in DARPA KAIROS (RESIN-11, NAACL 2022 Demo, first author).